

$$Y = Y_1, \dots, Y_n \sim N(\mu, \sigma^2)$$

μ, σ^2 are unknown

$$H_0: \mu = \mu_0$$

$$H_a: \mu > \mu_0$$

$$\Theta = (\mu, \sigma^2)$$

$$\Omega_0 = \{(\mu_0, \sigma^2): \mu = \mu_0, \sigma^2 > 0\}$$

Null Hypothesis

$$\Omega_a = \{(\mu, \sigma^2): \mu > \mu_0, \sigma^2 > 0\}$$

$$\Omega = \Omega_0 \cup \Omega_a = \{(\mu, \sigma^2): \mu \geq \mu_0, \sigma^2 > 0\}$$

$$L(\Omega) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}$$

$$\Lambda = \frac{L(\hat{\Omega}_0)}{L(\hat{\Omega})}$$

$$\hat{\Omega}_0 = \arg \max_{\mu, \sigma^2 \in \Omega_0} L(\Omega_0)$$

$$\hat{\Omega} = \arg \max_{\mu, \sigma^2 \in \Omega} L(\Omega)$$

Find the
 $\hat{\Omega}_0$

$$\Omega_0 = \{(\mu, \sigma^2) : \mu = \mu_0, \sigma^2 > 0\}$$

We are looking for $\hat{\mu}_0$ $\hat{\sigma}_0^2$
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indicate they belong to Ω_0

$$\hat{\mu}_0 = \mu_0$$

$$\hat{\sigma}_0^2 = ?$$

$$L(\sigma^2) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2}$$

Take derivative w.r.t. σ^2

$$\text{Set} = 0$$

Solve for σ^2

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mu)^2$$

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \mu_0)^2$$

$$\begin{aligned} \hat{\mu} &\rightarrow \mu \\ \hat{\mu} &= \mu_0 \end{aligned}$$

$$L(\hat{\Omega}_0) = L(\hat{\mu}_0, \hat{\sigma}_0^2) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\hat{\sigma}_0^2}\right)^{n/2} \times$$

$$\times \exp\left[-\frac{1}{2\hat{\sigma}_0^2} \sum (y_i - \mu_0)^2\right]$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\hat{\sigma}_0^2}\right)^{n/2} e^{-\frac{n}{2}}$$

$$L(\Omega) \quad \Omega = \{(\mu, \sigma^2) : \mu \geq \mu_0, \sigma^2 > 0\}$$

$$L(\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\sigma^2}\right)^{n/2} e^{-\frac{1}{2\sigma^2} \sum (y_i - \mu)^2}$$

$$\hat{\mu} = \text{① Take a log. of Likelihood}$$

$$\text{② Take derivative wrt } \mu$$

$$\text{③ Set to 0}$$

$$\text{④ Solve for } \mu$$

$$\hat{\mu} = \begin{cases} \bar{y} & \bar{y} > \mu_0 \\ \mu_0 & \bar{y} = \mu_0 \end{cases}$$

$$\hat{\sigma}^2 = \begin{cases} \frac{1}{n} \sum (y_i - \bar{y})^2 & \bar{y} > \mu_0 \\ \frac{1}{n} \sum (y_i - \mu_0)^2 & \bar{y} = \mu_0 \end{cases}$$

$$L(\hat{\Omega}) = L(\hat{\mu}, \hat{\sigma}^2) =$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\hat{\sigma}^2}\right)^{n/2} e^{-\frac{1}{2\hat{\sigma}^2} \sum (y_i - \bar{y})^2}$$

$$= \left(\frac{1}{\sqrt{2\pi}}\right)^n \left(\frac{1}{\hat{\sigma}^2}\right)^{n/2} e^{-\frac{n}{2}}$$

$$1 = \frac{L(\hat{\mu}_0)}{L(\hat{\mu})}$$

$$= \frac{\left(\frac{1}{\sqrt{n\pi}}\right)^n \left(\frac{1}{\hat{\sigma}_0^2}\right)^{n/2} e^{-n/2}}{\left(\frac{1}{\sqrt{n\pi}}\right)^n \left(\frac{1}{\hat{\sigma}_1^2}\right)^{n/2} e^{-n/2}}$$

$$1 = \begin{cases} \left(\frac{\hat{\sigma}_1^2}{\hat{\sigma}_0^2}\right)^{n/2} & \bar{y} > \mu_0 \\ 1 & \bar{y} = \mu_0 \end{cases}$$

$$X = \{X_1, \dots, X_n\} \stackrel{iid}{\sim} \text{Pois}(\theta_1)$$

$$Y = \{Y_1, \dots, Y_n\} \stackrel{iid}{\sim} \text{Pois}(\theta_2)$$

$$X \perp Y$$

$$H_0: \theta_1 = \theta_2 = \theta$$

$$H_a: \theta_1 \neq \theta_2$$

$$L(\theta_1, \theta_2) = \prod_{i=1}^n f(x_i) \prod_{j=1}^n f(y_j)$$

$$= \prod_{i=1}^n \frac{1}{x_i!} e^{-\theta_1} \theta_1^{x_i} \prod_{j=1}^n \frac{1}{y_j!} e^{-\theta_2} \theta_2^{y_j}$$

$$\Omega = \Omega_0 \cup \Omega_a$$

$$\Omega_0 = \{(\theta_1, \theta_2) : \theta_1 = \theta_2 = \theta\}$$

$$\Omega_a = \{(\theta_1, \theta_2) : \theta_1 \neq \theta_2, \theta_1 > 0, \theta_2 > 0\}$$

$$L(\Omega_0) = \prod_{i=1}^n \frac{1}{x_i!} e^{-\theta} \theta^{x_i} \prod_{j=1}^n \frac{1}{y_j!} e^{-\theta} \theta^{y_j}$$

$$k_x = \prod_{i=1}^n \frac{1}{x_i!}$$

$$k_y = \prod_{j=1}^n \frac{1}{y_j!}$$

$$e^{-\sum_{i=1}^n \theta - \sum_{j=1}^n \theta}$$

$$\theta^{\sum x_i + \sum y_j}$$

$$L(\Omega_0) = k_x k_y e^{-2n\theta} \theta^{n\bar{x} + n\bar{y}}$$

$$\hat{\Omega} = \hat{\Theta}$$

$$\ell(\Omega_0) = -2n\theta + (n\bar{x} + n\bar{y}) \ln(\theta) + \ln(k_x k_y)$$

$$\ell'(\Omega_0) = -2n + \frac{n\bar{x} + n\bar{y}}{\theta} = 0$$

$$\hat{\Theta} = \frac{1}{2}(\bar{x} + \bar{y})$$

$$L(\Omega) = k_x e^{-n\theta_1} \theta_1^{n\bar{x}} k_y e^{-n\theta_2} \theta_2^{n\bar{y}}$$

$$\ell(\Omega) = -n\theta_1 + n\bar{x} \ln \theta_1 - n\theta_2 + n\bar{y} \ln \theta_2 + \dots$$

$$\hat{\Theta}_1 = \bar{x} \quad \hat{\Theta}_2 = \bar{y} \quad \hat{\Theta} = \frac{\bar{x} + \bar{y}}{2}$$

$$L(\hat{\Omega}_0) = k_x k_y e^{-2n\hat{\Theta}} \hat{\Theta}^{n\bar{x} + n\bar{y}}$$

$$L(\hat{\Omega}) = k_x k_y e^{-n\bar{x} - n\bar{y}} \bar{x}^{n\bar{x}} \bar{y}^{n\bar{y}}$$

$$\Lambda = \frac{\left(\frac{1}{2}(\bar{x} + \bar{y})\right)^{n\bar{x} + n\bar{y}}}{\bar{x}^{n\bar{x}} \bar{y}^{n\bar{y}}}$$